Project Progress 1-ENSEMBLE OF GENERATIVE AND DISCRIMINATIVE TECHNIQUES FOR SENTIMENT ANALYSIS OF MOVIE REVIEWS

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Introduction

- Sentiment analysis predicts whether a text (such as a movie review) is positive or negative.
- It is typically treated as a binary classification problem.
- Simple machine learning models can already perform well.
- However, challenges remain:
  - Representing long documents
  - Capturing word order
  - Improving accuracy beyond simple bag-of-words methods

Main Idea of the paper

- Instead of relying on one model, the authors:
  - Combine multiple approaches
  - Combine different models into an ensemble
  - Show that combining models performs better than any single model

How does the method work(High level)

The paper combines:

1. Discriminative models
    - Linear classifiers(e.g., NB-SVM)
    - Focus directly on classification
2. Generative models
    - Train one language model on positive reviews
    - Train another language model on negative reviews
    - Compare which model is more likely to generate a test review

How Prediction is Made: If a review is more likely to be generated by the positive model, then classify it as positive

- For a new review:
    - Compute probability from the positive model
    - Compute probability from the negative model
- Use the likelihood ratio
- Combine results from all models
- Final decision is based on weighted combination

Models used:

- N-gram language models
- RNN language models

Linear Classification (NB-SVM)

What approach are we using?

- Use bag-of-words and n-grams as features
- Apply supervised reweighting (NB-SVM)

What NB-SVM Does

- Computes log-ratio of word frequencies:
  - Positive documents
  - Negative documents
- Uses this as input to:
  - Logistic regression or
  - Linear SVM

Improvement

- Adding trigrams improved performance
- Achieved strong baseline results (~91%)

Sentence Vectors

Main Idea

- Learn vector representations of:
  - Words
  - Paragraphs

How It Works

- Predict nearby words
- Learn embeddings that capture meaning
- Represent each document as a fixed-size vector

Why is this useful?

- Captures semantic meaning
- Goes beyond simple word counts

Model Ensemble

Core Idea:

- Instead of relying on a single model to make predictions, the authors combine multiple models to make the final sentiment prediction as a way to mitigate the weaknesses of each.

Each model analyzes movie reviews in a different way:

- The generative model estimates how likely a review is to be positive or negative.
- The NB-SVM model uses weighted word features for classification.
- The sentence vector model captures overall semantic meaning.

How the Ensemble works

1. Each model produces its own prediction score.
2. These scores are converted into probabilities.
3. The probabilities are combined using a weighted average.
4. The final sentiment is based on the combined score.

The weights are chosen using validation data to ensure the best performance.

The reason this works is because:

- Different models capture different types of information.
- Some models focus on word patterns.
- Others focus on overall language structure.
- Combining them reduces the weaknesses of any one model.
- This leads to more stable and more accurate predictions.

Results

Individual Model Performance

- N-gram Language Model = ~86.5%
- RNN Language Model = ~86.6%
- Sentence Vectors = ~88.7%
- NB-SVM (Trigrams) = ~91.9%

Ensemble Performance

- RNN + NB-SVM = ~92.1%
- Sentence Vectors + NB-SVM = ~92.4%
- All models combined = 92.57%

Key Takeaway

- The ensemble outperforms every individual model.
- Combining complementary approaches improves performance.
- Achieved new state-of-the-art accuracy on the IMDb dataset at the time.

Related work 1- BERT with Bidirectional LSTM. (Intro)

Main Idea

- Fine-tune BERT for movie review sentiment analysis.
- Add a BiLSTM layer on top of BERT to better capture long-range context.
- Apply the model to both:
  - Binary classification (positive/negative)
  - Fine-grained classification (multi-class sentiment)

Key Contributions

- Combines BERT + BiLSTM for improved accuracy.
- Uses oversampling and data augmentation to handle class imbalance.
- Computes overall sentiment polarity from the output vector.
- Reports performance competitive with or better than SOTA models.

Why It Matters

- Represents a modern deep learning approach.
- Shows how transformer-based models evolved beyond traditional methods.
- Provides comparison to classical ensemble approaches.

Related work

Traditional Sentiment Analysis

- Lexicon-based and machine learning methods (NB, KNN, Random Forest).
- Achieved ~81–90% accuracy on IMDb.
- Ensemble approaches improved results (~90%+).

Deep Learning Approaches

- CNNs and RNNs applied to sentiment analysis.
- LSTM and BiLSTM improved context understanding.
- Achieved ~87–90% accuracy on IMDb.

BERT-Based Approaches

- Fine-tuned BERT for binary classification (~92–93%).
- Applied to SST-2 and SST-5 for fine-grained classification.
- Transfer learning significantly improves performance.

Overall Sentiment Polarity Algorithm

Why not just use majority vote to classify?

If 51 reviews say negative and 49 say positive, then a simple majority says negative but that difference is not convincing enough. So the authors use threshold based rules.

- Binary classification(Algorithm 2):
  - If positive >= 1.2 * negative, then overall is positive.
- Else if:
  - Negative >= 1.2 * positive, then the overall is negative.
- Else:
  - The overall is neutral/mixed.

Algorithm 1: Overall sentiment polarity computation from three-class classification output vector.

Result: Dominating sentiment polarity for all reviews.

```
if #total neutral reviews > 85% of the total reviews then
    overall polarity ← neutral;
else
    if #total positive reviews >1.5 × # of total negative reviews then
        overall polarity ← positive;
    else if #total negative reviews >1.5 × # of total positive reviews then
        overall polarity ← negative;
    else
        overall polarity ← neutral;
```

Algorithm 2: Overall sentiment polarity computation from binary classification output vector.

Result: Dominating sentiment polarity for all reviews.

```
if #total positive reviews >1.2 × # of total negative reviews then
    overall polarity ← positive
else if #total negative reviews >1.2 × # of total positive reviews then
    overall polarity ← negative
else
    overall polarity ← neutral
```

The three class classification uses a similar model. For 3 classes positive, negative, and neutral.

If Neutral is 85% of all the reviews, then overall is neutral. Else, uses the Binary classification but the multiplier is 1.5 * class.

The reason this works is because this makes it less likely that border line cases are sorted inaccurately and this is more stable than just majority voting.

Overall Sentiment Polarity Algorithm(Continued)

Algorithm 3: Overall sentiment polarity computation from four-class classification output vector.

```
Result: Dominating sentiment polarity for all reviews.
if # (highly negative reviews + negative reviews) >1.2 × # of (positive reviews + highly positive reviews) then
    if #highly negative reviews >1.5 × # of negative reviews then
        overall polarity ← highly negative
    else
        overall polarity ← negative
else if #(positive reviews + highly positive reviews) >1.2 × # of (highly negative reviews + negative reviews) then
    if #highly positive reviews >1.5 × # of positive reviews then
        overall polarity ← highly positive
    else
        overall polarity ← positive
else
    overall polarity ← neutral
```

Algorithms 3 and 4 work the same way for four and five class.

Algorithm 3 works the same way as algorithm 2 but there is an extra conditional check to see if it is a highly positive/negative review or just positive/negative.

Algorithm 4 combines algorithm 3 with algorithm 2 neutral check.

Algorithm 4: Overall sentiment polarity computation from five-class classification output vector.

```
Result: Dominating sentiment polarity for all reviews.
if #total neutral reviews > 85% of the total reviews then
    overall polarity ← neutral
else
    if (#highly negative reviews + #negative reviews) >1.2 × # of (positive reviews + highly positive reviews) then
        if #highly negative reviews > #1.5 × negative reviews then
            overall polarity ← highly negative
        else
            overall polarity ← negative
    else if #(positive reviews + highly positive reviews) >1.2 × # of (highly negative reviews + negative reviews) then
        if #high positive reviews >1.5 × # of positive reviews then
            overall polarity ← highly positive
        else
            overall polarity ← positive
    else
        overall polarity ← neutral
```

Experiments - Datasets used

- IMDB
  - Binary
  - 3 class
  - 4 class
- SST
  - SST-2
  - SST-5 (5 class)

Model Setup

- Pre-trained bert-base-uncased.
- Added:
  - BiLSTM layer
  - Dense and softmax
- Evaluated using accuracy

The goal was to test the model performance across multiple datasets and classification complexity.

Results and Key findings

Binary classification:

- IMDb = 97.67%
- MR = 97.68%
- SST 2 = 97.62%
- Amazon 2 = 98.76%

TABLE II
ACCURACY (%) COMPARISONS OF MODELS ON BENCHMARK DATASETS FOR BINARY CLASSIFICATION

Model name	Dataset			
	IMDb-2	*MR*	*SST-2*	*Amazon-2*
RNN-Capsule [47]	84.12	83.80	82.77	82.68
coRNN [18]	87.4	87.11	88.97	89.32
TL-CNN [18]	87.70	81.5	87.70	88.12
Modified LMU [48]	93.20	93.15	93.10	93.67
DualCL [49]	-	94.31	94.91	94.98
L Mixed [50]	95.68	95.72	-	95.81
EFL [19]	96.10	96.90	96.90	96.91
NB-weighted-BON+dv-cosine [16]	97.40	-	96.55	97.55
SMART-RoBERTa Large [51]	96.34	97.5	96.61	-
Ours	**97.67**	**97.88**	**97.62**	**98.76**

Fine-Grained classification:

- IMDb 3 = 81.87%
- IMDb 4 = 70.75%
- SST 5 = 60.48%
- Amazon 5 = 69.68%

TABLE III
ACCURACY (%) COMPARISONS FOR THREE AND FOUR CLASS CLASSIFICATION ON IMDD

Model name	Dataset	
	IMDb-3	*IMDb-4*
CNN-RNF-LSTM [52]	73.71	63.78
DPCNN [22]	76.24	66.17
BERT-large [10]	77.21	66.87
Ours	**81.87**	**70.75**

TABLE VI
OVERALL POLARITY COMPUTATION ON ALL THE DATASETS

Dataset	Original OP	Computed OP[b]
IMDb-2	Neutral	Neutral
IMDb-3	Neutral	Neutral
IMDb-4	Neutral	Neutral
MR reviews	Neutral	Neutral
SST-2	Neutral	Neutral
SST-5	Neutral	Neutral
Amazon-2	Positive	Positive
Amazon-5	Positive	Positive

TABLE IV
ACCURACY (%) COMPORISONS OF MODELS ON BENCHMARK DATASETS FOR FIVE CLASS CLASSIFICATION

Model name	Dataset	
	SST-5	*Amazon-5*
CNN+word2vec [23]	46.4	48.85
TL-CNN [18]	47.2	58.1
DRNN [53]	-	64.43
BERT-large [10]	55.5	65.83
BCN+Suffix+BiLSTM-Tied+Cove [54]	56.2	65.92
RoBERTa+large+Self-explaining [55]	59.10	-
Ours	**60.48**	**69.68**

As we can see, this model performs extremely well with binary classification but the more classes are added, the lower the accuracy is. However, we can see that the model stays consistent with different types of datasets. Text augmentation (NLPAUG) improved fine-grained results.

TABLE V
ACCURACY (%) OF OUR MODEL WITH ACCURACY IMPROVEMENT TECHNIQUES ON SST-5

Classification task	Accuracy
BERT+BiLSTM	58.44
BERT+BiLSTM+SMOTE	58.36
BERT+BiLSTM+NLPAUG	**60.48**